

Vigtige regler vdr. tællelige mængder

Sætning 1.3

- Hvis A er en **delmængde** af en **tællelig** mængde, så er A **tællelig**.
- Hvis A **indeholder** en **overtællelig** mængde, så er A **overtællelig**.

Sætning 1.4: 'tællelig forening af tællelige mængder er tællelig'

Hvis $A_1, A_2, A_3, \dots$ er **tællelige** mængder, så er $A = A_1 \cup A_2 \cup A_3 \cup \dots$ **tællelig**.

Bevisidé Sætning 1.4: Cantor's diagonalisering

a_{11} a_{12} a_{13} a_{14} $\dots$

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$$\begin{array}{cccccc} a_{11} & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & a_{22} & a_{23} & a_{24} & \cdots \end{array}$$

Bevisidé Sætning 1.4: Cantor's diagonalisering

$$\begin{array}{ccccccccc} a_{11} & a_{12} & a_{13} & a_{14} & \cdots & & & & \\ a_{21} & a_{22} & a_{23} & a_{24} & \cdots & & & & \\ a_{31} & a_{32} & a_{33} & a_{34} & \cdots & & & & \end{array}$$

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$$\begin{array}{cccccc} a_{11} & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & a_{22} & a_{23} & a_{24} & \cdots \\ a_{31} & a_{32} & a_{33} & a_{34} & \cdots \\ a_{41} & a_{42} & a_{43} & a_{44} & \cdots \\ a_{51} & a_{52} & a_{53} & a_{54} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{array}$$

Bevisidé Sætning 1.4: Cantor's diagonalisering

$$\begin{array}{ccccccccc} a_{11} & \rightarrow & a_{12} & a_{13} & a_{14} & \cdots & & & \\ a_{21} & & a_{22} & a_{23} & a_{24} & \cdots & & & \\ a_{31} & & a_{32} & a_{33} & a_{34} & \cdots & & & \\ a_{41} & & a_{42} & a_{43} & a_{44} & \cdots & & & \\ a_{51} & & a_{52} & a_{53} & a_{54} & \cdots & & & \\ \vdots & & \vdots & \vdots & \vdots & & & & \end{array}$$

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$$\begin{array}{ccccccccc} a_{11} & \rightarrow & a_{12} & a_{13} & a_{14} & \cdots & & & \\ & \swarrow & & & & & & & \\ a_{21} & & a_{22} & a_{23} & a_{24} & \cdots & & & \\ & & & & & & & & \\ a_{31} & & a_{32} & a_{33} & a_{34} & \cdots & & & \\ & & & & & & & & \\ a_{41} & & a_{42} & a_{43} & a_{44} & \cdots & & & \\ & & & & & & & & \\ a_{51} & & a_{52} & a_{53} & a_{54} & \cdots & & & \\ \vdots & & \vdots & \vdots & \vdots & & & & \end{array}$$

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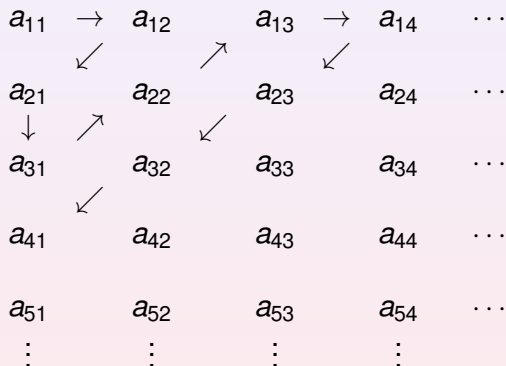
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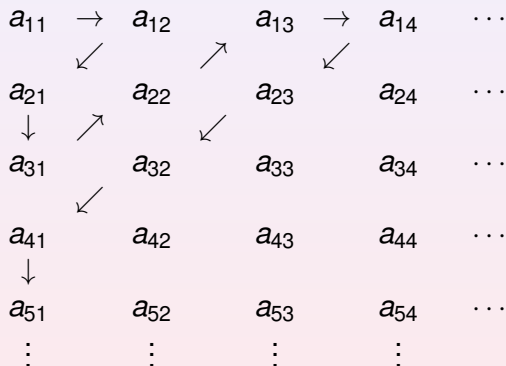
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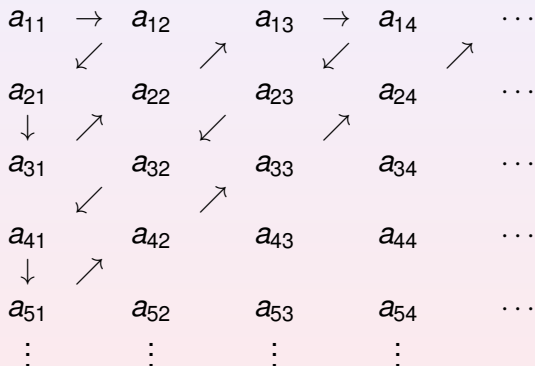
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$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\dots$
$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\dots$
$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\dots$
$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\dots$
$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

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$\frac{1}{1}$	$\rightarrow$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\dots$
	$\swarrow$				
$\frac{2}{1}$		$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\dots$
$\frac{3}{1}$		$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\dots$
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$\vdots$		$\vdots$	$\vdots$	$\vdots$	

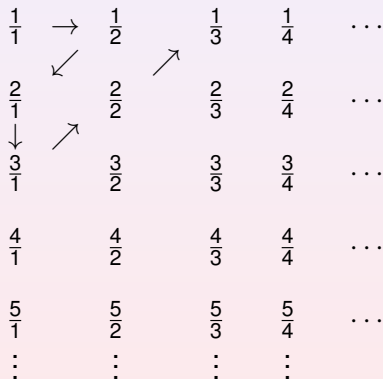
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	$\swarrow$				
$\frac{2}{1}$		$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\dots$
$\downarrow$					
$\frac{3}{1}$		$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\dots$
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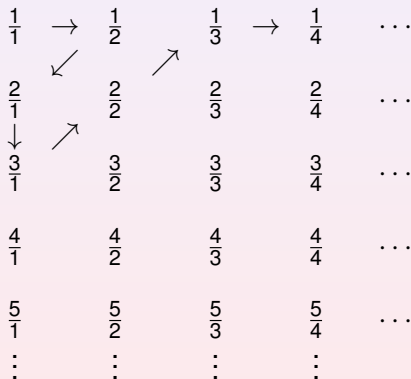
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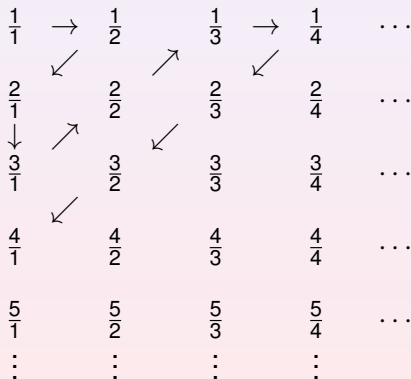
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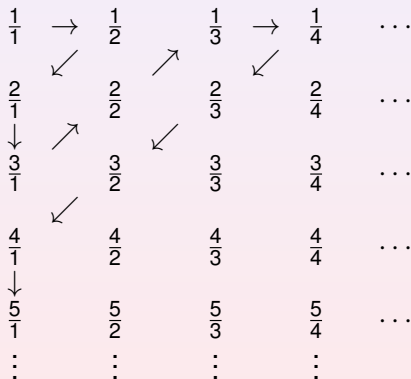
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